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AIDC ESS

Reference Architectures and Early Orders Point to ESS Adoption

Siemens (SIE GR, OW, covered by Phil Buller) announced that it is working with **NVIDIA** (NVDA US, OW, Covered by Harlan Sur) and partnering with **Fluence Energy** (FLNC US, Neutral, covered by Mark Strouse) to develop reference power-and-controls architecture for AIDCs. Within this architecture, Fluence Energy contributes the battery energy storage system. Together with Fluence Energy's previous announcement on MAS with hyperscalers, we believe these should rebuff market concerns on whether AIDCs need ESS. During our recent visit to **Sungrow**, management shared a similar view that ESS is a major solution to AIDC power constraints and they have already received ESS orders from data-center power developers. Meanwhile, **LEGS** (373220 KS, OW, covered by Parsley Ong) also announced that it has signed a \$1.6bn contract to supply 6GWh of ESS batteries to DTE Energy. This is in line with our key takeaways from the expert call we hosted ([link](#)), where the expert saw that ESS is likely to be part of the AIDC power solution in a tight power supply market, but orders will likely be more back-end loaded, due to the development sequence. Within the APAC region, we highlight key OW-rated names under J.P. Morgan APAC coverage: CATL-H/A, Sungrow, Samsung SDI, LGES and Nari.

Siemens partners with Fluence Energy to develop power architecture for AIDCs: Siemens, working with NVIDIA and **partnering with Fluence Energy**, has developed UL-aligned reference power-and-controls architecture for next-generation AIDCs that translates the "AI factory" concept into a deployable, modular electrical blueprint (official release [link](#)). Within this architecture, **Fluence Energy contributes battery energy storage** to add resilience and flexibility in power-constrained environments – supporting capabilities such as ride-through, black start, demand response, and AI load smoothing – so operators can bring extreme-density compute online faster while staying within fixed power envelopes.

Key takeaways from the AIDC ESS developments: We have seen a few positive developments with respect to AIDC ESS. In addition to the one mentioned above, in early May 2026, Fluence Energy announced that it had signed Master Supply Agreements (MAS) with two hyperscalers and expected first orders during F3Q (report [link](#)). We believe these developments should rebuff market concerns on whether AIDCs need ESS and result in rising expectations for AIDC ESS demand into FY26/27.

How this aligns with APAC companies: During our visit to Sungrow (report [link](#)) in mid May 2026, management shared that ESS is a major solution to AIDC power constraints; Sungrow has already received ESS orders from data-center power developers, while these ESS will be used to support AIDCs. This should further reinforce the thesis for AIDC ESS deployment. In late May 2026, LG Energy Solution (373220 KS, OW, covered by Parsley Ong) announced that it had signed a \$1.6bn contract to supply 6GWh of ESS batteries to DTE Energy (see Parsley's report [link](#)).

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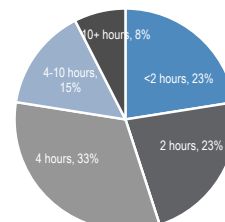
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Re-cap from the AIDC ESS expert call: We hosted an expert call on AIDC ESS (report [link](#)) in late April 2026, where an industry expert shared that ESS is likely to be part of the AIDC power solution in a tight power supply market, as most new AIDCs will require BESS for reliability and optimization, with rising attachment rates and durations. However, AIDC ESS will likely be more back-end loaded, due to the development sequence. Furthermore, distinct AIDC ESS requirements favour leading players with strong product quality, i.e. Sungrow and LGES.

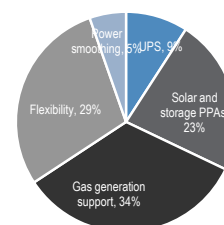
- **ESS adoption in AIDCs appears slow, development sequence explains the lag:** AIDC ESS adoption is taking time due to construction sequencing, not a lack of demand. ESS installation in a typical AIDC build occurs in the later stages — after land and civil works, foundation works, onsite power generation and the construction of data center halls and control buildings. Most large-scale AIDCs are still in the early construction phases. ESS procurement and installation should accelerate meaningfully as these projects progress toward completion. The expert's view is that the best period for ESS demand is still ahead, concentrated in the back half of the current build cycle.
- **Distinct AIDC ESS requirements favor players with strong product quality:** AIDC ESS is a distinct product category from utility-scale ESS, despite similar battery hardware. Utility-scale ESS is optimized for energy arbitrage, daily cycling and predictable loads. In contrast, AIDC ESS is designed for power quality, redundancy and the protection of valuable IT equipment, requiring rapid response, high power density and frequent micro-cycling. This demands high-quality product. The industry is moving toward integrating BESS and UPS into a unified energy layer.

Mix of launches of BESS in data centers, by storage duration



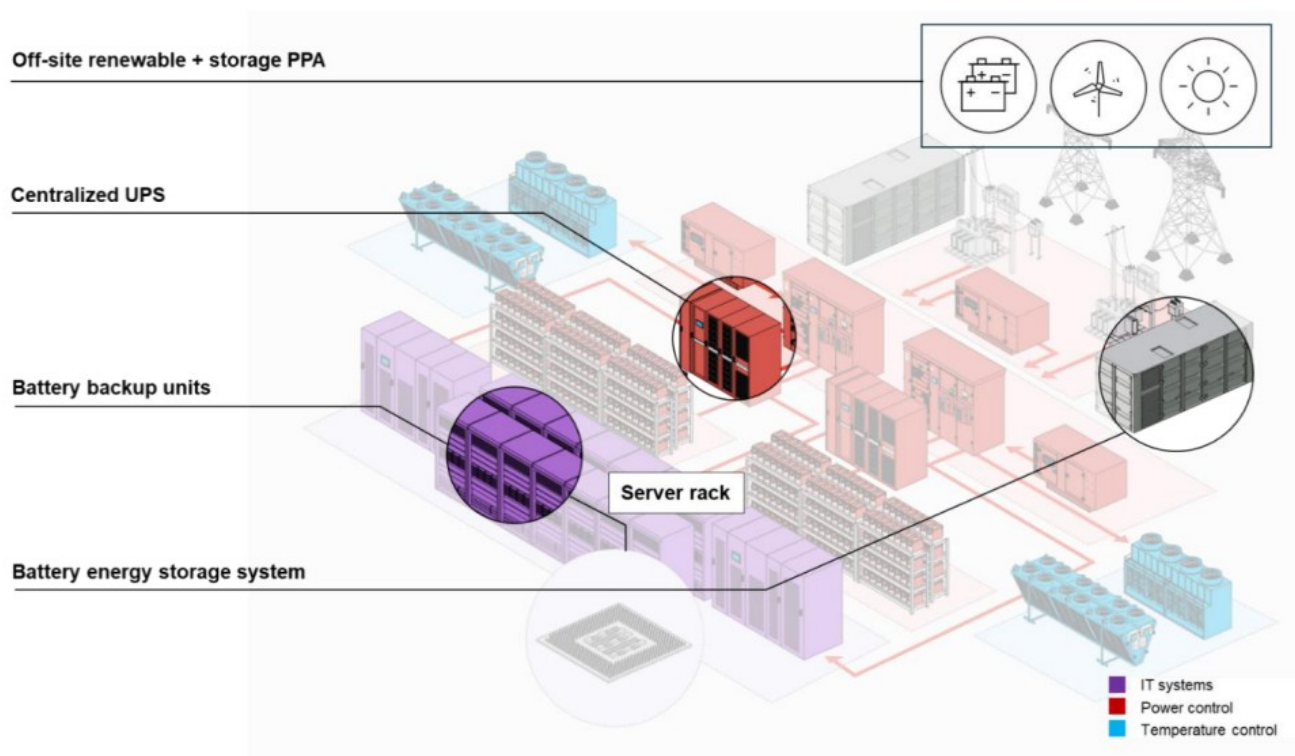
Source: BloombergNEF. Note: Not all projects disclose storage duration. Data as of 30 March 2026.

Mix of AIDC ESS use cases (in terms of capacity) in the US, averaged over 2026E-30E



Source: BloombergNEF.

Figure 1: Levels of energy storage in a data center



Source: BloombergNEF. PPA = power purchasing agreement; UPS = uninterruptible power supply.

Figure 2: Types of energy storage used in data centers

	Off-site energy storage	UPS	BBU	On-site energy storage
Adoption	Common	Near universal	Hyperscale/AI	Limited
Battery chemistry	Lithium-ion (LFP) Redox-flow, iron-air	Lead-acid Lithium-ion (NMC, LFP) Nickel-zinc	Lithium-ion (LFP) Supercapacitor	Lithium-ion (LFP) Redox-flow, iron-air
Runtime	2-4 hours 5+ hour storage less common	5-30 mins	1.5-5 mins	2-4 hours 5+ hour storage less common
Purpose	Co-located with renewable power generation	Short-term back up/ride through to generator Power smoothing	Short term back up/ride through to generator Power smoothing	Supporting on-site generation Flexibility Power smoothing
Grid interactive/flexible	Yes	Can be but usually not	No	Yes
Ownership	Energy developer	Data center	Rack owner	Energy developer, alternative models

Source: BloombergNEF. Note: UPS = uninterruptible power supply; NMC = nickel manganese cobalt; LFP = lithium iron phosphate; BBU = battery backup unit.

Recommendations

The table below summarizes the key relevant stocks under J.P. Morgan coverage and their exposure to the ESS business.

- **We recently initiated coverage of Deye at OW.** Deye is a distributed generation (DG) energy storage (ESS) producer with a first-mover advantage in Emerging Markets (EM). We see structural growth in ESS globally, on energy transition and renewables' intermittent issues. ESS battery shipment data in April also indicated DG has higher growth (C&I +154%yoy) than utility-scale (+93% yoy). By geography, multiple EM regions will see >50% 2025-27 installation CAGR, per Bloomberg New Energy Finance (BNEF). Near term, energy supply shocks are catalyzing faster adoption of DG of solar +storage, especially in regions where DG diesel was adopted. Longer term, we may see a policy tilt towards energy resilience, providing a long-term growth tailwind. Deye's legacy exposure to home appliances enables its cost leadership, owing to the rise in in-house fabrication ([link](#) to initiation report).
- **Largest PCS supplier – Sungrow (OW):** Sungrow is the world's largest solar inverter producer by volume, gaining market share in recent years due to cost advantage. Sungrow offers superior product quality and stronger brand recognition compared to domestic peers. We expect Sungrow to benefit from robust ESS demand and rising orders from utilities and data centers.
- **China ESS battery: CATL-A (OW)** is the largest ESS battery maker globally and our top pick in China's battery value chain. **EVE Energy (N)** has the highest ESS exposure among covered battery makers, followed by **CALB (N)**. We also favor **BYD-H/A (OW)**, given strong demand for its EV models in overseas markets and benefits from vertical integration in EV and ESS.

- Korean battery players see improved opportunities and margins in US ESS:** ESS order win momentum at **LGES (OW)** has been strong, and we estimate that the company has the largest LFP ESS order backlog among Korean battery makers. ESS visibility is also improving for **SDI (OW)**: US battery import data indicate that China-origin shipments have fallen further since July 2025 (post-OBBBA), while Korea-origin (largely SDI) shipments have risen sequentially, supporting an ongoing sourcing shift as US regulations remain effective. SDI's LFP cathode contract with L&F implies ~25GWh/year of US ESS shipments by 2028-29.

Figure 3: Summary of ESS value chain stocks under J.P. Morgan coverage

Company	Ticker	Cover Analyst	JPM Rating	ESS business	ESS revenue exposure			ESS volume exposure		
					FY25A	FY26E	FY27E	FY25A	FY26E	FY27E
CATL	300750 CH	Rebecca Wen	OW	ESS battery	15%	18%	22%	18%	22%	26%
CALB	3931 HK	Rebecca Wen	N	ESS battery	32%	27%	30%	41%	40%	43%
EVE Energy*	300014 CH	Rebecca Wen	N	ESS battery	40%	38%	40%	59%	54%	56%
Gotion	002074 CH	Rebecca Wen	UW	ESS battery	19%	21%	24%	30%	33%	37%
BYD-H	1211 HK	Nick Lai	OW	ESS battery	4%	5%	n/a	20%	27%	n/a
LGES	373220 KS	Parsley Ong	OW	ESS battery	13%	33%	41%	8%	23%	31%
Samsung SDI	006400 KS	Sonny Lee	OW	ESS battery	22%	30%	31%	33%	44%	49%
Sungrow	300274 CH	Alan Hon	OW	PCS+EMS+integration	42%	49%	53%	n/a	n/a	n/a
Deye	605117 CH	Alan Hon	OW	PCS+battery+packs	74%	86%	89%	n/a	n/a	n/a
Nari	600406 CH	Stephen Tsui	OW	BMS+PCS+EMS+integration	n/a	n/a	n/a	n/a	n/a	n/a
Xuji	000400 CH	Stephen Tsui	OW	PCS+EPC	n/a	n/a	n/a	n/a	n/a	n/a

Source: Company data, J.P. Morgan estimates. Note: ESS battery may include BMS or system integration. EVE's ESS volume exposure refers to ESS battery volume over total power battery, not incl. consumer battery. Note: Gotion FY25 exposure is JPM. Note: SDI's total revenue includes electronic materials and small battery while volume excludes electronic materials and small battery.

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A(605117.SS/Rmb111.10/OW), EVE Energy(300014.SZ/Rmb65.48/N), LG Energy Solution(373220.KS/W443,500/OW), Nari
Technology - A(600406.SS/Rmb24.64/OW), Samsung SDI(006400.KS/W601,000/OW), Sungrow - A(300274.SZ/Rmb172.70/
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LG Energy Solution (373220.KS, 373220 KS) Price Chart



Date	Rating	Price (W)	Price Target (W)
26-Oct-23	OW	409500	570,000
25-Jan-24	OW	380500	475,000
23-Apr-24	OW	378500	430,000
28-Jul-24	N	325000	330,000
08-Oct-24	N	420000	370,000
09-Nov-24	N	398500	340,000
09-Jul-25	OW	307000	360,000
27-Jul-25	OW	363500	420,000
27-Oct-25	OW	486500	580,000
24-Dec-25	OW	390000	500,000
09-Jan-26	OW	369500	430,000
08-Feb-26	OW	389500	450,000
22-Apr-26	OW	488000	510,000
30-Apr-26	OW	473000	530,000

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Apr 21, 2022. All share prices are as of market close on the previous business day.

Samsung SDI (006400.KS, 006400 KS) Price Chart



Date	Rating	Price (W)	Price Target (W)
28-Jul-23	OW	662000	1,000,000
19-Oct-23	OW	512000	850,000
12-Dec-23	OW	446000	600,000
31-Jan-24	OW	374500	550,000
31-Jul-24	OW	330500	400,000
07-Oct-24	OW	361500	450,000
16-Mar-25	N	191400	210,000
25-Apr-25	N	184700	200,000
29-Oct-25	N	309000	330,000
02-Mar-26	N	468000	400,000
30-Mar-26	OW	398500	650,000
24-Apr-26	OW	634000	720,000
28-Apr-26	OW	638000	770,000

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
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